Microeconomics I

Class 9 — Risk aversion measured; stochastic dominance; review

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Plan for today

Problem set 2 back; warm-up
Arrow–Pratt: how risk averse is she?
In pairs: r_A and r_R
Comparing risk aversion; insurance
First-order stochastic dominance (FOSD)
In pairs: two cumulative distribution functions (CDFs)
Second-order dominance; mean-preserving spreads
Break
Practice: Part I on one page; five problems in pairs
Summary; the Part I exam

Slide 1 Title

Preview version. The full slides and handout are posted after class.

Class 9 closes Part I. Class 8 defined risk aversion as concavity of the Bernoulli utility function and measured it in euros by the certainty equivalent; today that measurement becomes a number attached to each level of wealth, the Arrow–Pratt coefficient, which lets us compare two people and explain why insurance is bought. Then lotteries are compared: when everybody who likes money agrees, and when every risk averter agrees. The second half of the class is practice: the whole of Part I on one page and five problems in pairs. MWG 6.C–6.D.

Slide 2 Plan for today

Before the break, the theory of MWG 6.C–6.D in compact form: a warm-up on homogeneity from classes 4 to 6; the Arrow–Pratt coefficients and why the second derivative of u alone measures nothing, worked once and then practised in pairs; what it means for one person to be more risk averse than another; why a fair insurance price buys full cover; the distribution function of a money lottery and the first way in which one lottery beats another, practised in pairs; and the second way (same mean, less risk), with the mean-preserving spread. After the break, practice: Part I on one page and five problems in pairs, each tried first and then solved at the board.

MWG 6.C–6.D; Muñoz-Garcia chapter 5 is the gentler read.

Practicalities

- Problem set 2: returned in this class, graded
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu

Slide 3 Practicalities

Problem set 2 comes back today, graded, one copy per group; the marking is online behind the QR code on each group's cover, and questions about it go by e-mail. Nothing from today's class was on it.

Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu.

Warm-up: double every price (and, for x and v, wealth)

At $p = (2, 3)$, $w = 12$: $x = h(p, 6) = (3, 2)$, $v = 6$, $e(p, 6) = 12$.
Per pair, one minute: four values.

$x(4, 6, 24) = ?$

$v(4, 6, 24) = ?$

$h\big((4, 6), 6\big) = ?$

$e\big((4, 6), 6\big) = ?$

Class 4, slide 14; class 5 (v); class 6 (h, e). Mas-Colell, Whinston and Green (MWG) Def. 2.E.1, 3.D, 3.E.

Slide 4 Warm-up: double every price (and, for x and v, wealth)

Four values from classes 4 to 6 that you know at $p = (2, 3)$ and $w = 12$, asked again after every price and wealth have doubled. Before class, write the four values and say for each which property of x, v, h or e you used, and whether you needed the utility function at all.

Answers revealed in class. Then try the variant in which only the prices double and wealth stays at 12 euros, and ask what changes.

Arrow–Pratt: $r_A(x) = -u''(x)/u'(x)$ and $r_R(x) = x\, r_A(x)$

$$r_A(x) = -\frac{u''(x)}{u'(x)}$$ absolute risk aversion (risks in euros)

$$r_R(x) = -\frac{x\, u''(x)}{u'(x)} = x\, r_A(x)$$ relative risk aversion (risks as shares of wealth)

u: the Bernoulli utility over wealth x (class 8).

$u = \sqrt{x}$: $r_A(x) = \dfrac{1}{2x}$, $r_R(x) = \frac{1}{2}$

r_A falls with x. Class 8's two-coin game for Ana ($\sqrt{x}$; 64 euros if both coins land heads, else 0): risk premium 12 euros on the game alone, 3 euros with 36 sure euros more.

Why not u'' alone? $10\sqrt{x} + 3$ makes the same choices as $\sqrt{x}$ (class 8), yet has 10 times its u''. r_A and r_R are unchanged by any $\beta u + \gamma$, $\beta > 0$.

MWG Def. 6.C.3 and u'' alone, p. 190; Def. 6.C.4 (decreasing absolute risk aversion), p. 193; Def. 6.C.5, p. 194.

Slide 5 Arrow–Pratt: $r_A(x) = -u''(x)/u'(x)$ and $r_R(x) = x\, r_A(x)$

The coefficient of absolute risk aversion (MWG Def. 6.C.3, p. 190) is a function of wealth, not a single number: positive for a risk averter, zero for a risk-neutral person with linear u, negative for a risk lover, and it concerns risks of a fixed number of euros. Relative risk aversion (Def. 6.C.5, p. 194) multiplies by wealth and concerns risks that are a fixed share of wealth. For $\sqrt{x}$, $u' = \frac{1}{2}x^{-1/2}$ and $u'' = -\frac{1}{4}x^{-3/2}$, so $r_A(x) = 1/(2x)$ and $r_R(x) = \frac{1}{2}$ at every wealth.

Why not u'' alone: class 8 showed that $\beta u + \gamma$ with $\beta > 0$ is the same person, yet it multiplies u'' by β; dividing by u' cancels the factor, so r_A and r_R measure the person, not the units. The absolute coefficient of $\sqrt{x}$ falls as wealth rises, MWG's decreasing absolute risk aversion (Def. 6.C.4, p. 193): the grey line's two-coin game costs Ana a risk premium of 12 euros on its own, but only 3 euros once she holds 36 euros for sure.

In pairs: r_A, r_R and the attitude of four Bernoulli functions

Per pair: three answers per row ($x > 0$). Three minutes.

	$u(x)$	$r_A(x)$	$r_R(x)$	attitude
(a)	$4\sqrt{x} + 1$			
(b)	$-e^{-x/50}$			
(c)	$-\dfrac{1}{x^2}$			
(d)	x^2			

Answers revealed in class. Try it before class.

MWG Example 6.C.4 (constant r_A), p. 191; Def. 6.C.5, p. 194.

Slide 6 In pairs: r_A, r_R and the attitude of four Bernoulli functions

Four Bernoulli utility functions; per pair, the absolute and the relative coefficient of each and the attitude it describes, using the definitions of slide 5. Before class, differentiate twice and keep the minus sign; before computing, predict from the shape of each u what the sign of the coefficient should be, then check.

Answers revealed in class. One of the four rows is a rescaling of a function you already know; see whether you can tell which without computing.

More risk averse: r_A never lower, certainty equivalents never higher

u_2 is more risk averse than u_1 — three equivalent tests:

(i) $r_A(x, u_2) \geq r_A(x, u_1)$ for every x — at least as curved everywhere

(ii) $u_2 = \psi(u_1)$, ψ increasing and concave

(iii) $c(F, u_2) \leq c(F, u_1)$ for every F — in euros

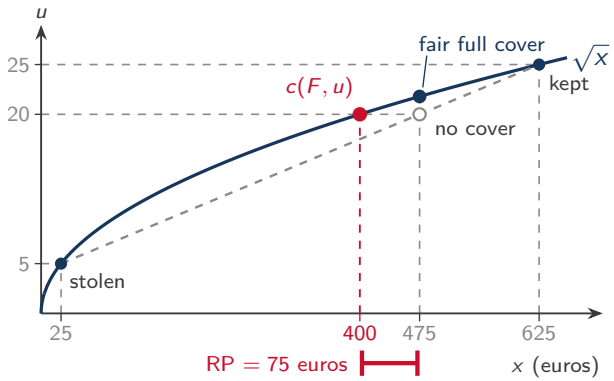
Certainty equivalent (class 8): the sure amount $c(F, u)$ with $u\big(c(F, u)\big) = \int u(x)\, dF(x)$.

A partial order: many pairs are not comparable.

Compare people by certainty equivalents, never by utility numbers.

MWG Prop. 6.C.2, p. 191: (i)–(v) are equivalent; (iv) and (v) are not needed here. Partial ordering, p. 192.

Insurance at a fair price: full cover



a 600-euro bike + 25 euros
stolen with probability $\pi = \frac{1}{4}$

no cover: 625 or 25 euros
$E = 475$, $c = 400$ euros;
risk premium $\mathrm{RP} = E - c = 75$ euros

fair cover, $q = \pi$ per euro:
600 euros of cover cost 150
475 euros for sure

At a fair price every strictly risk-averse buyer insures fully (cover = the 600-euro loss). She would pay up to 225 euros for an expected payout of 150 euros.

MWG Example 6.C.1, pp. 187–188; price above π: partial cover, p. 189.

Slide 7 More risk averse: r_A never lower, certainty equivalents never higher

MWG gives five ways of saying that one expected utility maximizer is more risk averse than another and proves that they agree (Prop. 6.C.2, p. 191); three are enough here. The first compares the coefficients at every wealth, the second says that u_2 is a concave transformation of u_1, and the third, a certainty equivalent no higher for every lottery, is the one that compares two people in euros; utility numbers themselves can never be compared across people, because each u can be rescaled, as on slide 5. The grey line recalls class 8's certainty equivalent: the sure amount whose utility equals the expected utility of the lottery.

The relation is only a partial order: for two functions whose coefficients cross at some wealth, neither person is more risk averse than the other.

Slide 8 Insurance at a fair price: full cover

MWG's Example 6.C.1 in round numbers. Each euro of cover costs q and pays one euro if the bike is stolen; she buys α euros of it, and full cover is $\alpha = D = 600$ euros, the whole loss. At a fair price, q equal to the probability of theft, expected wealth is 475 euros however much cover she buys, so she chooses among lotteries with the same mean, and a risk averter prefers the sure mean to any of them. Full cover is the one choice that delivers the sure mean, which is why the result holds for every strictly concave u and not only for $\sqrt{x}$: it is Jensen's inequality again, not a computation.

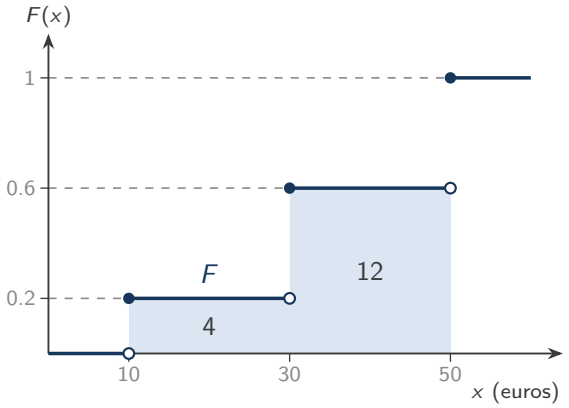
With $\sqrt{x}$ the uninsured position is worth 400 euros for sure, a risk premium (RP) of 75 euros, so she would pay up to $625 - 400 = 225$ euros for a cover whose expected payout is 150 euros; an insurer pooling many policies cares only about the average, so a premium above the expected loss can leave both sides better off. At a price above the probability of loss the model predicts partial cover (MWG p. 189).

Comparing lotteries: stochastic dominance

MWG 6.D

slide 9 of 21

A lottery's CDF: $F(x)$ = Prob(at most x); jumps are probabilities



F: 10, 30, 50 euros
with 0.2, 0.4, 0.4

$F(x) = \text{Prob}(\text{payoff} \leq x)$

jump at 30 euros: $0.6 - 0.2 = 0.4$

area under F up to 50 euros:
$4 + 12 = 16$ euros
$= 50 - E[x] = 50 - 34$

Equal means ⇔ equal areas under the two CDFs (from 0 to where both reach 1).

MWG 6.C, pp. 183–184, Fig. 6.C.1; (6.D.1), p. 198.

slide 10 of 21

Slide 9 Comparing lotteries: stochastic dominance

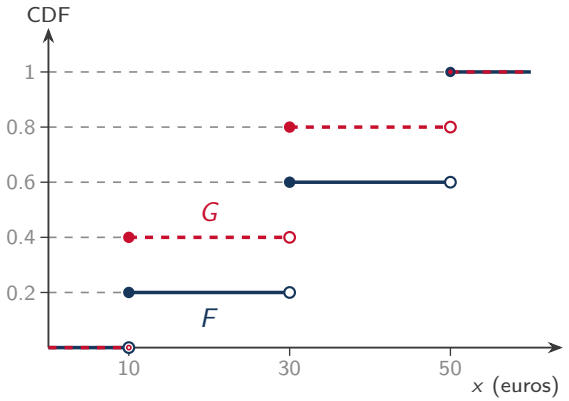
Part 2: so far we compared people; now two lotteries, and when everybody in a given class of people agrees on them. MWG 6.D.

Slide 10 A lottery's CDF: $F(x)$ = probability of at most x; jumps are probabilities

Class 8 drew the distribution function of a two-outcome lottery; the skill is to draw one for any lottery, so here it is for three outcomes. The function is a running sum: at each outcome it jumps by that outcome's probability and stays flat until the next one, and the filled dot sits at the value after the jump because "at most 30 euros" includes 30. The trap is to draw the probabilities themselves as bars, or a line through the points, instead of the running sum; a check is that the last jump reaches 1.

The area under the steps matters for slide 13. Up to the largest outcome it equals that outcome minus the mean, so for non-negative payoffs the area above F and below 1 is the expected value; two lotteries with the same mean therefore have the same total area under their distribution functions, which is what the second-order test uses.

F dominates G (first order): $F(x) \leq G(x)$ at every x



first-order stochastic dominance (FOSD)
F FOSD G: $\int u\,dF \geq \int u\,dG$ for every nondecreasing u

everyone who likes money agrees

$\Leftrightarrow$ $F(x) \leq G(x)$ for every x

on	$[10, 30)$	$[30, 50)$	$[50, \infty)$
F	0.2	0.6	1
G	0.4	0.8	1

F never above G: everyone who likes money weakly prefers F (means 34, 26 euros).

MWG Def. 6.D.1 and Prop. 6.D.1, p. 195; Fig. 6.D.1, p. 196. G: 10, 30, 50 euros with 0.4, 0.4, 0.2.

Slide 11 F dominates G (first order): $F(x) \leq G(x)$ at every x

First-order dominance is defined by agreement: every expected utility maximizer with a nondecreasing u likes F at least as much as G (MWG Def. 6.D.1, p. 195). The proposition turns that into a picture: the lower distribution function wins, because at every threshold the chance of ending at or below it is no larger; here F is G with probability mass pushed up, 0.2 from 10 to 30 euros and 0.2 from 30 to 50 euros (MWG Example 6.D.1, p. 196). The idea of the proof is a person who cares only about reaching a threshold, $u = 1$ above it and 0 below: if F were above G at that threshold, that person would prefer G.

Two warnings from MWG p. 196: dominance implies a higher mean (34 euros against 26) but a higher mean does not imply dominance, and dominance does not mean that every outcome of F beats every outcome of G (the outcomes here are the same three). One more: the inequality may hold with equality on some intervals, but once the two functions cross there is no dominance in either direction.

In pairs: two CDFs, P and Q

P: 10 or 50 euros, $\frac{1}{2}$ each Q: 20, 30, 60 euros with $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{4}$

Per pair, one graph: compare interval by interval. Does either dominate? Three minutes.

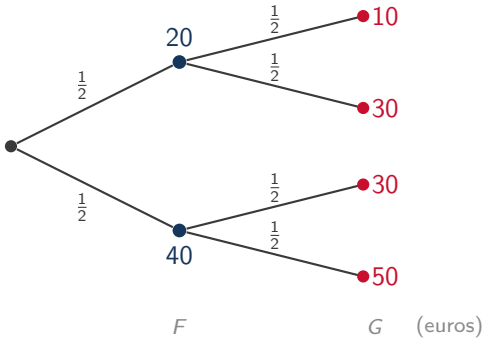
Answers revealed in class.
Try it before class.

Slide 12 In pairs: two CDFs, P and Q

Two lotteries, and the first-order test of slide 11 applied by hand: per pair, draw both distribution functions on one graph, then go interval by interval and say on which one is above the other, and conclude whether either dominates. Try it before class, and compute both means as well.

Answers revealed in class. Check that each function reaches 1 at its last jump, and put the filled dot at the value after each jump.

Same mean, more spread: every risk averter weakly prefers F



second-order stochastic dominance (SOSD)
F SOSD G: same mean, and $\int u\,dF \geq \int u\,dG$ for every nondecreasing concave u

mean-preserving spread: each outcome of F becomes a lottery with that mean

G: 10, 30, 50 euros with $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{4}$.
Both means: 30 euros.

G is a mean-preserving spread of F, so F SOSD G.
Equivalent test, same mean: $\int_0^x G(t)\,dt \geq \int_0^x F(t)\,dt$ at every x (areas under the CDFs).

MWG Def. 6.D.2, p. 197; Example 6.D.2, pp. 197–198; (6.D.2), p. 198; Prop. 6.D.2, p. 199.

Break

Ten minutes

Slide 13 Same mean, more spread: every risk averter weakly prefers F

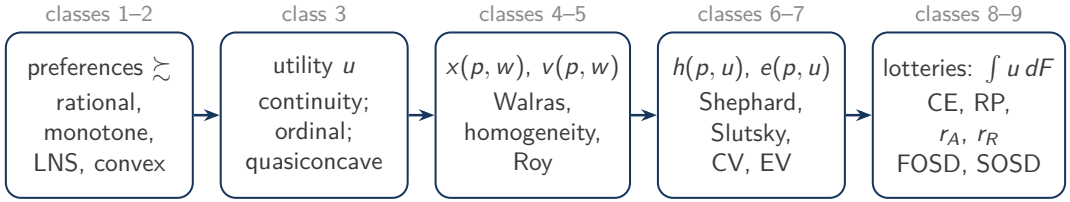
Second-order dominance is defined only for lotteries with the same mean, and again by agreement, now of a smaller group: everyone with a nondecreasing concave u (MWG Def. 6.D.2, p. 197). The tree is MWG's recipe for making a lottery riskier without changing its mean (Example 6.D.2, pp. 197–198): play F, then replace each outcome by a small lottery whose mean is that outcome. Every risk averter likes F at least as much because at each branch she compares a sure 20 euros with a lottery of mean 20 euros, and the same at 40 euros; that is Jensen's inequality from class 8, applied branch by branch, and with $\sqrt{x}$ the certainty equivalents are about 29.1 euros for F against 28.1 for G.

A written answer says that the means are equal and which outcome is spread into which sub-lottery; a larger variance alone is no proof. The takeaway's second line is the test for when nobody gives you the tree (MWG (6.D.2), Prop. 6.D.2): with equal means, F dominates G exactly when the area under G from 0 to any x is at least that under F; here the difference runs 0, 2.5, 0, 2.5, 0 at 10, 20, 30, 40 and 50 euros.

Slide 14 Break

Ten minutes.

Part I on one page: from preferences to uncertainty



LNS: locally non-satiated · CV, EV: compensating and equivalent variation
CE: certainty equivalent

Part I = classes 1–9. Now five problems in pairs: try first, then the board.

MWG 1.B, 2.C–2.E, 3.B–3.G, 3.I, 6.B–6.D.

Slide 15 Part I on one page: from preferences to uncertainty

The five problems that follow walk back through the boxes: demand, indirect utility, expenditure and Hicksian demand (classes 4 to 7); the Slutsky equation at one point (class 7); expected utility in the probability triangle (class 8); certainty equivalent and mean-preserving spread (classes 8 and 9); comparing risk aversion (class 9). The first two boxes, preferences and utility, were practised in classes 2 and 3 and in their exercises.

One thread runs across: under certainty any increasing transformation of u is the same person; under risk the Bernoulli u only up to $\beta u + \gamma$, $\beta > 0$, because a non-affine change can flip a choice between lotteries (class 8).

Demand, indirect utility, expenditure, Hicksian demand

A consumer has $u(x_1, x_2) = 2 \ln x_1 + x_2$, $x_1 > 0$.
Prices $p_1, p_2 > 0$, wealth $w > 2p_2$, so she buys both goods.
In pairs: one set of answers per pair.

(a) Walrasian demand $x(p, w)$. Answers revealed in class. Try it before class.

(b) Indirect utility $v(p, w)$.
(c) Expenditure function $e(p, u)$, by inverting v (for $u > 2 \ln(2p_2/p_1)$).
(d) $h_1(p, u)$ by Shephard's lemma; compare it with $x_1(p, w)$.

Classes 4–7: MWG 3.D, 3.E, 3.G (Shephard's lemma, Roy's identity).

Slide 16 Demand, indirect utility, expenditure, Hicksian demand

The pipeline of classes 4 to 7 on one consumer: Walrasian demand from the tangency, indirect utility, the expenditure function by inverting v in wealth, Hicksian demand from Shephard's lemma. Try the four parts in this order; write the tangency with the price ratio the right way up.

Answers revealed in class. Before (d), predict from the form of u whether compensating her should change her demand for good 1.

Substitution and wealth effects at one point

Ines's Walrasian demand for good 1 is $x_1(p, w) = w\,p_2 \,/\, (p_1\,(p_1 + p_2))$ (two goods).
At $p = (1, 3)$ and $w = 32$; $h_1(p, u)$: her Hicksian demand for good 1, at $u = v(p, w)$.
In pairs: one set of answers per pair.

(a) Her demand x_1, and $\partial x_1/\partial p_1$. Answers revealed in class. Try it before class.

(b) The wealth effect,
$-x_1\,\partial x_1/\partial w$.

(c) The substitution effect
$\partial h_1/\partial p_1$ (Slutsky).

(d) At this point, which falls faster
as p_1 rises, x_1 or h_1? Why?

Classes 6–7: MWG 3.G (Hicksian demand, the Slutsky equation).

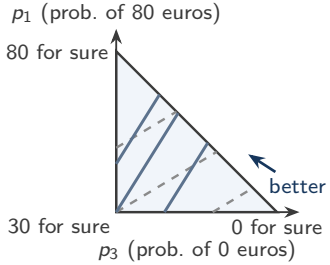
Slide 17 Substitution and wealth effects at one point

The Slutsky equation of class 7 at one price and wealth: everything is a number. Before class, write the equation, compute the pieces that Walrasian demand alone gives, and solve for the substitution effect; then say which of x_1 and h_1 reacts more to the price, and why.

Answers revealed in class. Keep every sign, and differentiate the quotient before you put the numbers in.

Indifference curves in the probability triangle

Prizes: 80, 30 and 0 euros. Petra maximizes expected utility. Solid: three of her indifference curves. Dashed: each line joins lotteries with the same expected value. Better lotteries lie up and to the left.



Which statement is true?
In pairs: a letter and a one-line reason.

A. She prefers 30 euros for sure to every other lottery with expected value 30 euros.
B. She prefers some lottery with expected value 30 euros to 30 euros for sure.
C. She is risk neutral: her curves are straight lines, like the dashed ones.
D. Nothing follows about her attitude to risk: her curves are not the dashed ones.

Answer revealed in class. Try it before class.

Class 8; MWG Fig. 6.B.5, p. 175; Def. 6.C.1, p. 185.

Slide 18 Indifference curves in the probability triangle

A multiple-choice item on class 8's probability triangle: her indifference curves and the lines of equal expected value are both drawn, and one statement about her attitude to risk is true. Before class, write down the definition of risk aversion, locate the sure amounts in the triangle, and prepare a one-line reason for your letter.

Answer revealed in class. Compare the two lines through the corner the options name.

Dani chooses between two bonuses

Bonus F: 8 or 64 euros, $\frac{1}{2}$ each. Bonus G: 0, 16 or 64 euros, with $\frac{1}{4}$, $\frac{1}{4}$, $\frac{1}{2}$.
Dani's Bernoulli utility: $u(x) = x^{1/3}$.
In pairs: one set of answers per pair.

(a) Dani's $c(F, u)$, and $\mathrm{RP} = E[x] - c(F, u)$. Answers revealed in class. Try it before class.

(b) Show that G is a mean-preserving spread of F.

(c) Use (b): which bonus does every risk averter prefer, and why?

MWG Def. 6.C.2, p. 186; Example 6.D.2 and Def. 6.D.2, pp. 197–198.

Slide 19 Dani chooses between two bonuses

The uncertainty skills of today and class 8 on one pair of bonuses: the certainty equivalent and risk premium by the three steps of class 8, the mean-preserving spread built with the means checked first, and the choice of every risk averter argued without computing. Try the three parts before class.

Answers revealed in class. Keep the units straight at every step, euros or utils.

Who is more risk averse, Marta or Nil?

Marta and Nil maximize expected utility over wealth $x > 0$ (euros).
Marta: $u_M(x) = \sqrt{x}$. Nil: $u_N(x) = -1/x$. In pairs: one set of answers per pair.

(a) $r_A(x)$ for Marta and for Nil. Answers revealed in class. Try it before class.

(b) Who is more risk averse? Check tests (i) and (ii) of slide 7.

(c) Wealth is 4 or 36 euros, probability $\frac{1}{2}$ each: Marta's and Nil's certainty equivalents.

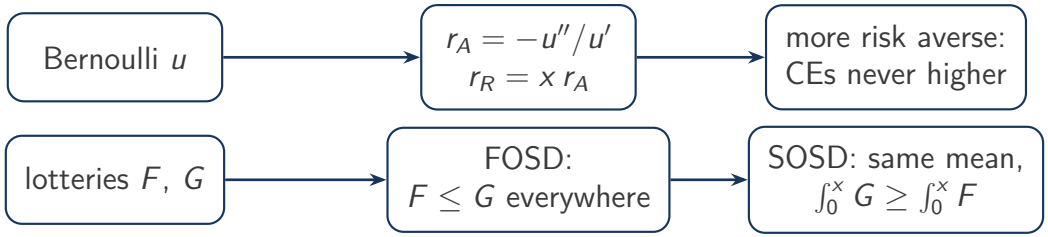
MWG Def. 6.C.2, p. 186; Def. 6.C.3, p. 190; Prop. 6.C.2, p. 191.

Slide 20 Who is more risk averse, Marta or Nil?

Today's three tests of slide 7 on one pair: the Arrow–Pratt coefficient from slide 5 for each, then tests (i) and (ii), then the certainty equivalent of one bonus for each by the three steps of class 8. Before class, compute both coefficients and find the ψ that turns one utility into the other.

Answers revealed in class. Certainty equivalents are in euros, whatever sign the utility numbers have.

Today in one line — and good luck on 28 October



Part I exam: Wednesday 28 October, Room 1030.

Arrive by 09:30; it starts at 09:35 sharp and ends at 11:20.

Bring photo ID and a pen. A simple calculator is allowed but never needed. Phones away.

Any handwritten notes, also as photocopies or printouts; no typed notes or other printed material.

Revise: Sextant, pkocourek.com/sextant/?c=micro-1

Part II (theory of the firm, general equilibrium) is taught by Olga Chiappinelli from November.

Thank you, and good luck.

Slide 21 Today in one line — and good luck on 28 October

The Arrow–Pratt coefficients measure how concave u is at each wealth and are unchanged by rescaling; one person is more risk averse than another when her coefficient is at least as high at every wealth, which is the same as a certainty equivalent no higher for every lottery; at a fair price a risk averter insures fully; one lottery first-order dominates another when its distribution function is nowhere above the other's, so that everyone who likes money agrees; with equal means it second-order dominates when the running area under the other's function is always at least as large, so that every risk averter agrees; that closes Part I, classes 1 to 9.

Part I exam: Wednesday 28 October, Room 1030. Arrive by 09:30; it starts at 09:35 sharp and ends at 11:20. Bring photo ID and a pen. Closed book; any handwritten notes are allowed, also as photocopies or printouts (nothing typed or otherwise printed, no formula sheet). A simple calculator is allowed but never needed; phones away.

Revise in Sextant: pkocourek.com/sextant/?c=micro-1. Part II (theory of the firm, general equilibrium) is taught by Olga Chiappinelli from November. Thank you, and good luck.